MYTHIC

Compiler Optimization Report - YOLOPX

June 2026

Document Version 3.0
SDK Version v26.05

MYTHIC

1 Summary

For YOLOPX at 736x1280x3 resolution:

Performance simulation estimates a Total Processing Time (for the compiled ONNX file) of

- 9.41 ms, of which
- 7.58 ms is Analog NPU processing time, which corresponds to
    - 72.6% ACE utilization, using 48 Analog Compute Engines; and of which
- 1.84 ms is Digital NPU processing time,
    - using only 1 Digital NPU core, which is sufficient to achieve 30 fps.

The number of used Digital NPU cores (1) was selected to achieve at least 30 fps. If more Digital NPU cores were used, the Digital NPU processing time could be substantially shortened. For example, 8 Digital NPU cores could reduce the Digital NPU processing time by ~8x, resulting in a total (Analog + Digital) processing time of ~7.8 ms, corresponding to ~130 fps.

There are no digital bottlenecks in the Analog NPU processing time.

Automatic optimization performs as well as the best manual optimization.

We may be able to improve performance in the future by finding a more optimal placement of weight blocks to ACEs (to increase the ACE utilization), but note that the current ACE utilization of 72.6% is already quite high.

We may be able to improve performance in the future by allowing weight block widths to exceed 256. The hardware design recently increased support to 512, and the compiler will add support for 512 in the future.

The estimated power is

- 0.7 Watts at the target frame rate (30 fps), and
- 2.2 Watts at the maximum frame rate (106 fps), if restricted to using only 1 Digital NPU core.

Power estimates are approximate, and will be more precise in the future.

2 Reproducing the results

The results in this report were produced using the `“high_optimization”` compiler config for YOLOPX.

Note that some variance in results is expected (up to a few percent), due to non-deterministic, multi-threaded compiler optimizations.

The `mythic-compiler` command was

```
mythic-compiler \
    --input-artifact $MYTHIC_SDK_ROOT/models/training/yolopx/yolopx_trained.tar.gz \
    --compiler-config $MYTHIC_SDK_ROOT/mythic-model-zoo/configs/yolopx/compiler/
yolopx_736x1280x3_m2048_high_optimization.yaml \
    --output-artifact /rscratch/tonbomythic3/<your_username>/
yolopx_736x1280x3_m2048_high_optimization_compiled.tar.gz
```

The `mythic-ppa-estimators` command for the maximum frame rate (106 fps) was

```
mythic-ppa-estimators \
    --estimate-performance \
    --estimate-power \
    /rscratch/tonbomythic3/<your_username>/
yolopx_736x1280x3_m2048_high_optimization_compiled.tar.gz
```

The `mythic-ppa-estimators` command for the target frame rate (30 fps) was

```
mythic-ppa-estimators \
    --estimate-performance \
    --estimate-power \
    --power-inference-rate 30 \
    /rscratch/tonbomythic3/<your_username>/
yolopx_736x1280x3_m2048_high_optimization_compiled.tar.gz
```

3 Performance estimates

The `mythic-ppa-estimators` tool outputs the following performance statistics:

```
Analog NPU Total Estimated Processing Time for Compiled ONNX File: 7.58 ms
Analog NPU Estimated Frame Rate for Compiled ONNX File: 131.95 fps (frames per second)
Critical Path ACE Latency: 7.58 ms
Maximum SRAM Read/Write Time (over 12 ACE tiles): 6.83 ms
Maximum SIMD Operation Time (over 12 ACE tiles): 0.00 ms
Average SRAM Read/Write Time (averaged over 12 ACE tiles): 5.42 ms
Average SIMD Operation Time (averaged over 12 ACE tiles): 0.00 ms
Total Executed ACE Operations: 1,649,376
Total Executed ACE MACs: 285,021,216,768
Maximum Theoretical ACE Execution Time (With No Parallelization Using 1 ACE): 263.90 ms
Minimum Theoretical ACE Execution Time (With Even Parallelization Across 48 ACEs): 5.50
 ms
ACE Utilization (Minimum Theoretical Time/Estimated Processing Time): 72.55%
Total SRAM Bytes Read Across Chip: 4,624,816,152 bytes
Total SRAM Bytes Written Across Chip: 2,401,140,480 bytes
Estimated Die Area to Achieve Estimated Processing Time: 253.09 mm^2
-------------------------------------------------------------------
Digital NPU Performance Metrics (from separate ONNX graph processing)
  Total Estimated MACs Targeting Digital NPU: 89,000,000
  Digital NPU MAC Utilization: 75.79%
  Digital NPU Cores: 1
  Digital NPU Frequency: 1,000 MHz
  Digital Estimated Frame Processing Time: 1.84 ms
  Digital Estimated Frame Rate: 544.84 fps (frames per second)
-------------------------------------------------------------------
```

```
Combined Analog + Digital NPU Latency: 9.41 ms
Combined Analog + Digital NPU Total Estimated Frame Rate: 106.23 fps (frames per second)

NOTE: The information reported above may include the Analog NPU or the Analog
NPU and a digital NPU as indicated. This estimated processing time and
associated frame rate only covers the execution of the compiled ONNX model(s)
running on one or both of those compute solutions in isolation. It does not
include PCIe transfer overhead. Performance estimation for additional processing
steps and a combined compute hardware solution will be added in future versions.
```

4 Power estimates

The `mythic-ppa-estimators` tool outputs the following power estimates.

Power at the target frame rate (30 fps):

```
Process Nodes:  Analog 28nm, Digital 5nm
Number of ACEs: 24
Number of Digital NPU Cores: 1
Inference Rate: 30 frames/second (fps)

Analog NPU Estimated Power: 0.677 W for target ONNX file
    Functional Unit Power: 0.528 W
    Interconnect Power:    0.149 W

Digital NPU Estimated Power: 0.002 W for digital target ONNX file

Total Combined (Analog + Digital NPU) Power: 0.679 W

NOTE: Estimation for leakage, clock tree, chip I/O power will be added in future
versions.
```

Power at the maximum frame rate (106 fps):

```
Process Nodes:  Analog 28nm, Digital 5nm
Number of ACEs: 24
Number of Digital NPU Cores: 1
Inference Rate: 106 frames/second (fps)

Analog NPU Estimated Power: 2.230 W for target ONNX file
    Functional Unit Power: 1.704 W
    Interconnect Power:    0.526 W

Digital NPU Estimated Power: 0.008 W for digital target ONNX file
```

```
Total Combined (Analog + Digital NPU) Power: 2.238 W

NOTE: Estimation for leakage, clock tree, chip I/O power will be added in future
versions.
```

5 Compiler reports

5.1 L0 IR stats

From near the end of mythic-compiler stdout:

```
================================= L0 IR stats ==================================
                   Tiles: 13
                 Buffers: 384
               Launchers: 629
           ACE Launchers: 257
             ACE Weights: 49858224
        ACE calculations: 1649376
    Max ACE calculations: 151248 on tile 2_3
================================================================================
```

5.2 ACE utilization report

The compiler-generated `ace_utilization.txt`:

```
Summary:

   - 74.52% ACE utilization is predicted by the compiler,

     - assuming no digital bottlenecks (which can typically be eliminated).

   - 1,649,376 ACE calculations are spread across 48 ACEs, which is

     - 34,362.00 ACE calculations per ACE (on average).

   - 46,111 ACE calculations are in the critical path, implying

     - 7.38 ms predicted inference time (160 ns per ACE calculation).

+------+------+--------------+--------------+-------------------+-------------+
|      |      |              |     ACE      |     Predicted     |  Predicted  |
|      |      |      ACE     | calculations |    Full Network   |     ACE     |
| Tile | ACEs | calculations |    Per ACE   | ACE Critical Path | Utilization |
+------+------+--------------+--------------+-------------------+-------------+
|  1_1 |    0 |          N/A |          N/A |               N/A |         N/A |
|      |      |              |              |                   |             |
|  3_1 |    4 |      134,274 |    33,568.50 |            46,111 |      72.80% |
|      |      |              |              |                   |             |
|  4_1 |    4 |      139,380 |    34,845.00 |            46,111 |      75.57% |
|      |      |              |              |                   |             |
|  1_2 |    4 |      139,840 |    34,960.00 |            46,111 |      75.82% |
|      |      |              |              |                   |             |
|  2_2 |    4 |      137,770 |    34,442.50 |            46,111 |      74.69% |
|      |      |              |              |                   |             |
|  3_2 |    4 |      136,804 |    34,201.00 |            46,111 |      74.17% |
|      |      |              |              |                   |             |
|  4_2 |    4 |      136,206 |    34,051.50 |            46,111 |      73.85% |
|      |      |              |              |                   |             |
|  1_3 |    4 |      136,942 |    34,235.50 |            46,111 |      74.25% |
|      |      |              |              |                   |             |
|  2_3 |    4 |      151,248 |    37,812.00 |            46,111 |      82.00% |
|      |      |              |              |                   |             |
|  3_3 |    4 |      136,850 |    34,212.50 |            46,111 |      74.20% |
|      |      |              |              |                   |             |
|  4_3 |    4 |      137,402 |    34,350.50 |            46,111 |      74.50% |
|      |      |              |              |                   |             |
|  1_4 |    4 |      135,240 |    33,810.00 |            46,111 |      73.32% |
|      |      |              |              |                   |             |
|  2_4 |    4 |      127,420 |    31,855.00 |            46,111 |      69.08% |
+------+------+--------------+--------------+-------------------+-------------+
```

```
|  All |   48 |    1,649,376 |    34,362.00 |              46,111 |      74.52% |
+------+------+--------------+--------------+--------------------+-------------+
```

5.3 Weight utilization report

The compiler-generated `weight_utilization.txt`:

```
Summary:

   - 83.42% of analog weight storage capacity is used, using

   - 49,858,224 weights out of 59,768,832 capacity, using

   - 257 weight blocks.

+------+--------+------------+------------+-------------+
|      | Weight |  Allocated |   Weight   |             |
| Tile | Blocks |   Weights  |  Capacity  | Utilization |
+------+--------+------------+------------+-------------+
|  3_1 |     22 |  4,040,672 |  4,980,736 |      81.13% |
|      |        |            |            |             |
|  4_1 |     20 |  4,329,088 |  4,980,736 |      86.92% |
|      |        |            |            |             |
|  1_2 |     22 |  4,414,240 |  4,980,736 |      88.63% |
|      |        |            |            |             |
|  2_2 |     22 |  4,340,288 |  4,980,736 |      87.14% |
|      |        |            |            |             |
|  3_2 |     22 |  4,111,616 |  4,980,736 |      82.55% |
|      |        |            |            |             |
|  4_2 |     22 |  3,882,240 |  4,980,736 |      77.95% |
|      |        |            |            |             |
|  1_3 |     23 |  4,387,536 |  4,980,736 |      88.09% |
|      |        |            |            |             |
|  2_3 |     20 |  3,942,480 |  4,980,736 |      79.15% |
|      |        |            |            |             |
|  3_3 |     22 |  3,890,688 |  4,980,736 |      78.11% |
|      |        |            |            |             |
|  4_3 |     21 |  4,197,760 |  4,980,736 |      84.28% |
|      |        |            |            |             |
|  1_4 |     20 |  4,301,248 |  4,980,736 |      86.36% |
|      |        |            |            |             |
|  2_4 |     21 |  4,020,368 |  4,980,736 |      80.72% |
+------+--------+------------+------------+-------------+
|  All |    257 | 49,858,224 | 59,768,832 |      83.42% |
+------+--------+------------+------------+-------------+
```

5.4 SRAM utilization report

The compiler-generated `sram_utilization.txt`:

```
Summary:

  - 32,096,422 bytes of SRAM is used, consisting of

    - 31,221,280 bytes for buffers (97.27%) and

    - 875,142 bytes for control flow (2.73%), using

  - 63.77% of the simulated on-chip SRAM capacity.

+------+------------+--------------+------------+------------+-------------+
|      |   Buffer   | Control Flow |   Total    |    SRAM    |             |
|      | Allocation |  Allocation  | Allocation |  Capacity  |             |
| Tile |  (bytes)   |   (bytes)    |  (bytes)   |  (bytes)   | Utilization |
+------+------------+--------------+------------+------------+-------------+
|  3_1 |  3,982,656 |       69,117 |  4,051,773 |  4,194,304 |      96.60% |
|      |            |              |            |            |             |
|  4_1 |    579,584 |       83,182 |    662,766 |  4,194,304 |      15.80% |
|      |            |              |            |            |             |
|  1_2 |  3,986,464 |       62,716 |  4,049,180 |  4,194,304 |      96.54% |
|      |            |              |            |            |             |
|  2_2 |  3,959,456 |       74,271 |  4,033,727 |  4,194,304 |      96.17% |
|      |            |              |            |            |             |
|  3_2 |  2,159,808 |       83,431 |  2,243,239 |  4,194,304 |      53.48% |
|      |            |              |            |            |             |
|  4_2 |    588,704 |       69,397 |    658,101 |  4,194,304 |      15.69% |
|      |            |              |            |            |             |
|  1_3 |  1,145,536 |       96,047 |  1,241,583 |  4,194,304 |      29.60% |
|      |            |              |            |            |             |
|  2_3 |  3,729,792 |       89,247 |  3,819,039 |  4,194,304 |      91.05% |
|      |            |              |            |            |             |
|  3_3 |  2,998,400 |       80,098 |  3,078,498 |  4,194,304 |      73.40% |
|      |            |              |            |            |             |
|  4_3 |    250,624 |       46,369 |    296,993 |  4,194,304 |       7.08% |
|      |            |              |            |            |             |
|  1_4 |  3,862,016 |       58,577 |  3,920,593 |  4,194,304 |      93.47% |
|      |            |              |            |            |             |
|  2_4 |  3,978,240 |       62,690 |  4,040,930 |  4,194,304 |      96.34% |
+------+------------+--------------+------------+------------+-------------+
|  All | 31,221,280 |      875,142 | 32,096,422 | 50,331,648 |      63.77% |
+------+------------+--------------+------------+------------+-------------+
```